

# Series and Convergence

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## 1 Series as limits of partial sums

Given a real sequence  $(a_n)_{n \geq 0}$ , the symbol  $\sum_{n=0}^{\infty} a_n$  denotes the formal infinite sum. Its value, when defined, is the limit of the sequence of partial sums.

**Definition 1** (Series and partial sums). The  $n$ -th partial sum of  $(a_n)$  is

$$s_n = \sum_{k=0}^n a_k.$$

The series  $\sum_{n=0}^{\infty} a_n$  is the symbolic object associated with  $(a_n)$ ; we say it *converges to*  $S \in \mathbb{R}$  iff  $s_n \rightarrow S$  as  $n \rightarrow \infty$ , and write  $\sum_{n=0}^{\infty} a_n = S$ . Otherwise the series *diverges*.

**Definition 2** (Absolute and conditional convergence). The series  $\sum a_n$  *converges absolutely* iff the series of non-negative terms  $\sum |a_n|$  converges. If  $\sum a_n$  converges but  $\sum |a_n|$  diverges, the convergence is *conditional*.

It is a standard fact that absolute convergence implies convergence: the partial sums of  $\sum a_n$  form a Cauchy sequence, dominated by the Cauchy tails of  $\sum |a_n|$ . The converse fails, as witnessed by  $\sum (-1)^{n+1}/n = \ln 2$  paired with the divergent harmonic series.

**Example 3** (Geometric series). For  $r \in \mathbb{R}$  with  $|r| < 1$ ,  $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$ . Indeed  $s_n = (1 - r^{n+1})/(1 - r) \rightarrow 1/(1 - r)$  as  $r^{n+1} \rightarrow 0$ . This is the workhorse comparator for the ratio test below.

## 2 The ratio test

The ratio test reduces convergence of  $\sum a_n$  to comparison with a geometric series.

**Theorem 4** (Ratio test). Let  $(a_n)$  be a sequence of nonzero real numbers and suppose

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

exists in  $[0, +\infty]$ . Then:

1. if  $L < 1$ , the series  $\sum_{n=0}^{\infty} a_n$  converges absolutely;
2. if  $L > 1$  (including  $L = +\infty$ ), the series diverges;
3. if  $L = 1$ , the test is inconclusive.

*Proof.* We treat the three cases.

**Case  $L < 1$ .** Choose  $r$  with  $L < r < 1$ . Since  $|a_{n+1}/a_n| \rightarrow L < r$ , there exists  $N \in \mathbb{N}$  such that

$$\left| \frac{a_{n+1}}{a_n} \right| < r \quad \text{for all } n \geq N.$$

Multiplying these inequalities for  $n = N, N+1, \dots, N+k-1$  gives

$$|a_{N+k}| < r^k |a_N| \quad \text{for every } k \geq 0.$$

Therefore, for all  $n \geq N$ ,

$$\sum_{m=N}^n |a_m| \leq |a_N| \sum_{k=0}^{n-N} r^k \leq \frac{|a_N|}{1-r}.$$

The partial sums of the non-negative series  $\sum |a_m|$  are bounded above, hence converge (a monotone bounded sequence has a limit). Adding back the finitely many terms for  $m < N$  shows  $\sum |a_n| < \infty$ , so  $\sum a_n$  converges absolutely.

**Case  $L > 1$ .** Choose  $r$  with  $1 < r < L$  (or  $r > 1$  arbitrary if  $L = +\infty$ ). There exists  $N$  such that  $|a_{n+1}/a_n| > r$  for all  $n \geq N$ , hence  $|a_{N+k}| > r^k |a_N|$ . Since  $r > 1$ ,  $|a_n| \rightarrow \infty$ , so  $a_n \not\rightarrow 0$  and the series diverges by the term test.

**Case  $L = 1$ .** The series  $\sum 1/n$  has  $L = 1$  and diverges, while  $\sum 1/n^2$  has  $L = 1$  and converges. No conclusion can be drawn from  $L$  alone.  $\square$

**Example 5** (Exponential series). For every  $x \in \mathbb{R}$ , setting  $a_n = x^n/n!$  gives

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{|x|}{n+1} \rightarrow 0.$$

Thus  $L = 0 < 1$ , and Theorem 4 yields absolute convergence of  $\sum_{n=0}^{\infty} x^n/n! = e^x$  for every real  $x$ . The same argument works for  $x \in \mathbb{C}$  and, with  $|x|$  replaced by  $\|A\|$ , for any square matrix  $A$ .

### 3 Power series

A *power series* centred at  $x_0$  is a series of the form  $\sum_{n=0}^{\infty} c_n(x - x_0)^n$ . Its set of convergence is an interval (possibly a single point or all of  $\mathbb{R}$ ) of *radius of convergence*

$$R = \frac{1}{\limsup_{n \rightarrow \infty} |c_n|^{1/n}} \quad (\text{Cauchy-Hadamard}).$$

Inside  $|x - x_0| < R$  the series converges absolutely; outside it diverges; behaviour at the endpoints requires a separate test.

This perspective is the bridge from analysis to ML: every smooth activation function, the matrix exponential, and the leading-order expansion of the KL divergence are all instances of power-series approximation.